

2. Conformal field theory

This chapter we develop necessary ideas & techniques from the world-sheet
qft: operator product expansion, conformal invariance, the Virasoro algebra,
and vertex operators. Focus: conformally invariant field theory in 2D (flat)

2.1 Massless scalars in two dimensions

Example of D free scalar fields in two dimensions, $X^\mu(\sigma^1, \sigma^2)$

We refer to these two dimensions as the world sheet.

The action is
$$S = \frac{1}{4\pi\alpha'} \int d^2\sigma \left(\partial_1 X^\mu \partial_1 X_\mu + \partial_2 X^\mu \partial_2 X_\mu \right) \quad (2.1.1)$$

This is the Polyakov action from previous chapter (eq (1.2.13) of book),
but with Euclidean metric instead of World sheet (γ_{ab}) metric.

↑ Euclidean convention (A.1.31). $\Rightarrow$ Most string calculations done in Euclidean
World-sheet. Results of this chapter apply to Minkowski world-sheet, and in
first 7 sections we can just replace σ^2 with $i\sigma^0$ to get Minkowski equations.
the index μ is still for flat Minkowski.

Eq (2.1.1) can be straightforwardly quantized canonically to get spectrum, vacuum
expectation values & so on. (We've essentially done this in Chapter 1 in light cone gauge)
(Not in these notes since they were preview of results that were quickly derived.)

We first develop various local properties such as equations of motions, operator product,
Ward identities, and conformal invariance, before working our way around to the
spectrum. It will be efficient for us to use the Path integral formalism.

Path integral representation will primarily be used to derive operator equations; these
can also be derived in Hilbert space formalism.

$\Rightarrow$ Q: How do these differ i.e. what is Path integral representation vs
Hilbert space formalism.

We adopt complex coordinates: $z = \sigma^1 + i\sigma^2$, $\bar{z} = \sigma^1 - i\sigma^2$
(bar for complex conjugate unless expression is longer where star is used).

Define:

$$\partial_z = \frac{1}{2}(\partial_1 - i\partial_2), \quad \partial_{\bar{z}} = \frac{1}{2}(\partial_1 + i\partial_2)$$

$$\Rightarrow \partial_z z = 1, \quad \partial_z \bar{z} = 0, \quad \partial_{\bar{z}} z = 0, \quad \partial_{\bar{z}} \bar{z} = 1$$

Can also write $\partial_z \equiv \partial$ or $\partial_{\bar{z}} \equiv \bar{\partial}$ when no ambiguity involved.

For a general vector V^a in the same way:

$$V^z = V^1 + iV^2, \quad V^{\bar{z}} = V^1 - iV^2, \quad V_z = \frac{1}{2}(V^1 - iV^2), \quad V_{\bar{z}} = \frac{1}{2}(V^1 + iV^2).$$

For indices 1, 2, the metric is identity & upper & lower is not distinguished.

For complex indices we get the raising & lowering by

$$g_{z\bar{z}} = g_{\bar{z}z} = \frac{1}{2}, \quad g_{zz}, g_{\bar{z}\bar{z}} = 0, \quad g^{z\bar{z}} = g^{\bar{z}z} = 2, \quad g^{zz} = g^{\bar{z}\bar{z}} = 0$$

Note: $d^2z = 2d\sigma^1 d\sigma^2$. We also define $\int d^2z \delta^2(z, \bar{z}) = 1$ so that $\delta^2(z, \bar{z}) = \frac{1}{2} \delta(\sigma^1) \delta(\sigma^2)$.

Divergence theorem in complex coordinates: $\int_R d^2z (\partial_z V^{\bar{z}} + \partial_{\bar{z}} V^z) = i \oint_{\partial R} (V^z d\bar{z} - V^{\bar{z}} dz)$, where the contour integral circles the region R counterclockwise.

In this notation the action is

$$S = \frac{1}{2\pi\alpha'} \int d^2z \partial X^\mu \bar{\partial} X_\mu$$

and the classical equation of motion is

$$\partial \bar{\partial} X^\mu(z, \bar{z}) = 0$$

We use notation $X^\mu(z, \bar{z})$ (even if a bit redundant) to reserve $f(z)$ (~~$f(z)$~~) for fields whose equation of motion makes them analytic (equivalently holomorphic) functions of z . Writing the equation of motion as $\partial(\bar{\partial} X^\mu) = \bar{\partial}(\partial X^\mu) = 0$ it follows that ∂X^μ is holomorphic and $\bar{\partial} X^\mu$ is antiholomorphic (holomorphic in $\bar{z}$), hence the notation $\partial X^\mu(z)$ and $\bar{\partial} X^\mu(\bar{z})$.

Under the Minkowski continuation $\sigma^2 = i\sigma^0$, a holomorphic field becomes a function only of $\sigma^0 - \sigma^1$ and an antiholomorphic field a function only of $\sigma^0 + \sigma^1$. We thus use as synonyms

holomorphic = left-moving,
antiholomorphic = right-moving.

Expectation values are defined by the path integral

$$\langle F[X] \rangle = \int [dX] \exp(-S) F[X],$$

where $F[X]$ is any functional of X , such as a product of local operators.

We do not normalize $\langle F[X] \rangle$ by dividing by $\langle 1 \rangle$.

The path integral of total derivative is zero:

- true for Bosonic path integrals (limit of infinite number of ordinary integrals)
- true for more formal path integral ~~with~~ with Grassmann variables for example as

$$\begin{aligned}
\Rightarrow 0 &= \int [dX] \frac{\delta}{\delta X_\mu(z, \bar{z})} \exp(-S) \\
&= \int [dX] \exp(-S) \frac{\delta S}{\delta X_\mu(z, \bar{z})} = - \left\langle \frac{\delta S}{\delta X_\mu(z, \bar{z})} \right\rangle \\
&= \frac{1}{\pi \alpha'} \left\langle \partial \bar{\partial} X^\mu(z, \bar{z}) \right\rangle
\end{aligned}$$

Same goes when we have arbitrary insertions not at z : $\left\langle \partial \bar{\partial} X^\mu(z, \bar{z}) \dots \right\rangle = 0$ (2.1.16)

$\Rightarrow$ We can regard the additional insertions as preparing arbitrary initial and final states (or we could do the same thing with boundary conditions, note: I don't understand what is meant by this).

$\Rightarrow$ (2.1.16) is the same as the statement in the Hilbert space formalism that $\partial \bar{\partial} \hat{X}^\mu(z, \bar{z}) = 0$ (from classical EOM (Ehrenfest's theorem: $\Rightarrow$ Operator Eq.))

holds for all matrix elements of the operator $\hat{X}^\mu(z, \bar{z})$. Thus we refer to relations that hold in the sense (2.1.16) as operator equations.

Above the "... " were for insertions located away from z .

When the insertion might be coincident with z , we get:

$$\begin{aligned}
0 &= \int [dX] \frac{\delta}{\delta X_\mu(z, \bar{z})} \left[\exp(-S) X^\nu(z', \bar{z}') \right] \\
&= \int [dX] \exp(-S) \left[\eta^{\mu\nu} \delta^2(z-z', \bar{z}-\bar{z}') + \frac{1}{\pi \alpha'} \partial_z \partial_{\bar{z}} X^\mu(z, \bar{z}) X^\nu(z', \bar{z}') \right] \\
&= \eta^{\mu\nu} \left\langle \delta^2(z-z', \bar{z}-\bar{z}') \right\rangle + \frac{1}{\pi \alpha'} \partial_z \partial_{\bar{z}} \left\langle X^\mu(z, \bar{z}) X^\nu(z', \bar{z}') \right\rangle
\end{aligned}$$

$\Rightarrow$ Equations of motion hold except at coincident points.

Again this holds with arbitrary additional insertions '...' in Path Integral as long as they are not in $(z, \bar{z})$ or $(z', \bar{z}')$:

$$\frac{1}{\pi \alpha'} \partial_z \partial_{\bar{z}} \left\langle X^\mu(z, \bar{z}) X^\nu(z', \bar{z}') \dots \right\rangle = - \eta^{\mu\nu} \left\langle \delta^2(z-z', \bar{z}-\bar{z}') \dots \right\rangle$$

$$\text{Thus } \frac{1}{\pi \alpha'} \partial_z \partial_{\bar{z}} X^\mu(z, \bar{z}) X^\nu(z', \bar{z}') = - \eta^{\mu\nu} \delta^2(z-z', \bar{z}-\bar{z}') \quad (2.1.20)$$

holds as an operator equation. (In the Hilbert space formalism, the product in the path integral becomes a time-ordered product, and the delta function comes from the derivatives acting on time ordering.)

In free field theory, it is useful to introduce the operation of normal ordering. The normal ordered operators, $:A:$, are defined as

$$:X^\mu(z, \bar{z}): = X^\mu(z, \bar{z})$$

$$:X^\mu(z_1, \bar{z}_1)X^\nu(z_2, \bar{z}_2): = X^\mu(z_1, \bar{z}_1)X^\nu(z_2, \bar{z}_2) + \frac{\alpha'}{2} \eta^{\mu\nu} \ln |z_{12}|^2,$$

where $z_{ij} = z_i - z_j$. This normal ordering is related to normal ordering in terms of raising & lowering operators. From this definition we get the property

$$\partial_1 \bar{\partial}_1 :X^\mu(z_1, \bar{z}_1)X^\nu(z_2, \bar{z}_2): = 0 \quad (2.1.23)$$

This follows from (2.1.20) & the equation $\partial \bar{\partial} \ln |z|^2 = 2\pi \delta^2(z, \bar{z})$

2.2 The operator product expansion

Object of interest in string perturbation theory: path integral $\langle \text{expectation value} \rangle$ of product of local operators,

$$\langle A_{i1}(z_1, \bar{z}_1) A_{i2}(z_2, \bar{z}_2) \dots A_{in}(z_n, \bar{z}_n) \rangle,$$

where A_i is some basis for the set of local operators. Important: to understand this $\langle EV \rangle$ in the limit where two operators are taken to approach one another.

$\Rightarrow$ For this we need a tool called Operator product expansion (OPE).

OPE says: a product of two local operators close together can be approximated to arbitrary accuracy by

$$A_i(\sigma_1) A_j(\sigma_2) = \sum_k C_{ij}^k(\sigma_1 - \sigma_2) A_k(\sigma_2).$$

This is operator statement, so $\langle A_i(\sigma_1) A_j(\sigma_2) \dots \rangle = \sum_k C_{ij}^k(\sigma_1 - \sigma_2) \langle A_k(\sigma_2) \dots \rangle$ as long as the separation between σ_1 & σ_2 is ~~small~~ small compared to the distance to other operators. The coefficient functions $C_{ij}^k(\sigma_1 - \sigma_2)$ depend i, j, k but not on the other operators. The terms are conventionally arranged in order of decreasing size in the limit as $\sigma_1 \rightarrow \sigma_2$. (Analogous to Taylor series $\sum$, but here coeff functions can be singular.)

OPE central to QFT

Aim of section: derivation of OPE for the X^μ theory, using special properties of free field theory.

Eq (2.1.23) says that the operator product is harmonic function of $(z, \bar{z})$. Harmonic function $\equiv$ A twice continuously differentiable function $f: U \rightarrow \mathbb{R}$, where U is an open subset of $\mathbb{R}^n$, that satisfies Laplace's equation - Wikipedia.

Theory of complex variables: A harmonic function is locally the sum of a holomorphic and an anti-holomorphic functions. $\Rightarrow$ Is nonsingular as $z_1 \rightarrow z_2$ and can be freely Taylor expanded in z_{12} and $\bar{z}_{12}$. Thus,

$$(2.2.4) \quad X^\mu(z_1, \bar{z}_1) X^\nu(z_2, \bar{z}_2) = -\frac{\alpha'}{2} \eta^{\mu\nu} \ln|z_{12}|^2 + :X^\nu X^\mu(z_2, \bar{z}_2): + \sum_{k=1}^{\infty} \frac{1}{k!} \left[(z_{12})^k :X^\nu \partial^k X^\mu(z_2, \bar{z}_2): + (\bar{z}_{12})^k :X^\nu \bar{\partial}^k X^\mu(z_2, \bar{z}_2): \right]$$

FOM means terms with mixed $\partial \bar{\partial}$ derivatives vanish

Eg (2.2.4) has ^{the} form of OPE. OPEs are usually used as asymptotic expansions, the first few terms giving the dominant behaviour at small separation.

The use of 'u' in place of '=' will mean 'equal up to nonsingular terms!'

(We often write OPEs as singular + unspecified nonsingular terms.)

OPEs are actually convergent in conformally invariant field theory

$\Rightarrow$ Can reconstruct the entire theory from the coefficient functions.

The various operators on the right-hand side of the OPE (2.2.4) involve product of fields at same point. Usually in qft such a product is divergent & must be cut off & renormalized, but here normal ordering makes it well defined.

Normal ordering: - convenient way to define operators in free field theory

- of little use in most interacting theories (additional divergences from vertex)

- However, we are interested in many free theories and ~~other~~ many others can be related to free ones.

$\Rightarrow$ Let's further define normal ordering for arbitrary number of fields:

$$:X^{\mu_1}(z_1, \bar{z}_1) \dots X^{\mu_n}(z_n, \bar{z}_n): = X^{\mu_1}(z_1, \bar{z}_1) \dots X^{\mu_n}(z_n, \bar{z}_n) + \sum \text{subtractions}, \quad (2.2.5)$$

where the sum runs over all permutations of pairs of fields with each permutation contributing a term of form: $\frac{1}{2} \alpha' \eta^{\mu_i \mu_j} \ln|z_{ij}|^2$

Example:

$$:X^{\mu_1}(z_1, \bar{z}_1) X^{\mu_2}(z_2, \bar{z}_2) X^{\mu_3}(z_3, \bar{z}_3): = X^{\mu_1}(z_1, \bar{z}_1) X^{\mu_2}(z_2, \bar{z}_2) X^{\mu_3}(z_3, \bar{z}_3) + \left(\frac{\alpha'}{2} \eta^{\mu_1 \mu_2} \ln|z_{12}|^2 X^{\mu_3}(z_3, \bar{z}_3) + \frac{\alpha'}{2} \eta^{\mu_1 \mu_3} \ln|z_{13}|^2 X^{\mu_2}(z_2, \bar{z}_2) + 1 \text{ permutation} \right).$$

This definition (2.2.5) satisfies the naive equation of motion (exercise: show it)

The definition can be compactly summarized as:

$$:F: = \exp\left(\frac{\alpha'}{4} \int d^2 z_1 d^2 z_2 \ln|z_{12}|^2 \frac{\delta}{\delta X^\mu(z_1, \bar{z}_1)} \frac{\delta}{\delta X_\mu(z_2, \bar{z}_2)}\right) :F:$$

$$\Rightarrow \mathcal{T} = : \mathcal{T} : + \sum \text{contractions},$$

where the contraction is the opposite of subtraction: $-\frac{1}{2} \alpha' \eta^{\mu, \nu} \ln |z_i|^2$

The OPE for any pair of operators can be generated from

$$:F::G: = :FG: + \sum \text{cross-contractions} \quad \text{for arbitrary functionals}$$

F and G of X . The sum runs over all ways of contracting pairs with one field in F and one in G . Also:

$$:F::G: = \exp\left(-\frac{\alpha'}{2} \int d^2 z_1 d^2 z_2 \ln |z_{12}|^2 \frac{\delta}{\delta X_F(z_1, \bar{z}_1)} \frac{\delta}{\delta X_G(z_2, \bar{z}_2)}\right) :FG: \quad (2.2.10)$$

where the functional derivatives only act on the fields in F or G respectively.

Example:

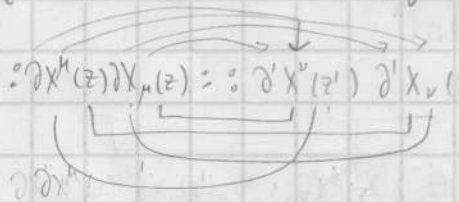
$$:\partial X^\mu(z) \partial X_\mu(z) : : \partial X^\nu(z') \partial X_\nu(z') :$$

$$= : \partial X^\mu(z) \partial X_\mu(z) \partial X^\nu(z') \partial X_\nu(z') : - 4 \cdot \frac{\alpha'}{2} (\partial \partial' \ln |z-z'|^2) : \partial X^\mu(z) \partial X_\mu(z') : \\ + 2 \cdot \eta^{\mu, \mu} \left(-\frac{\alpha'}{2} \partial \partial' \ln |z-z'|^2\right)^2$$

$$\sim \frac{D \alpha'^2}{2(z-z')^4} - \frac{2 \alpha'}{(z-z')^2} : \partial X^\mu(z') \partial X_\mu(z') : - \frac{2 \alpha'}{z-z'} : \partial'^2 X^\mu(z') \partial X_\mu(z') :$$

(Note to self: derive this in more detail)

The second term in the equality comes from the four ways of forming a single pair and the third from the two ways of forming two pairs.

$$:\partial X^\mu(z) \partial X_\mu(z) : : \partial X^\nu(z') \partial X_\nu(z') : \quad \begin{array}{l} (4\text{-way of single pair}) \\ (2\text{-way of double pair}) \end{array}$$


In the final line OPE is in standard form by Taylor expanding inside the normal ordering to express everything in terms of local operators z' and putting the most singular term first.

Another example:

$$F = e^{ik_1 \cdot X(z, \bar{z})}$$

$$G = e^{ik_2 \cdot X(0, 0)}$$

The variations $\delta/\delta X_F^\mu$ and $\delta/\delta X_G^\mu$ give factors of $ik_{1\mu}$ & $ik_{2\mu}$ respectively, so the general result (2.2.10) becomes

$$\begin{aligned} :e^{ik_1 \cdot X(z, \bar{z})}::e^{ik_2 \cdot X(0,0)}: &= \exp\left(\frac{\alpha'}{2} k_1 \cdot k_2 \ln|z|^2\right) :e^{ik_1 \cdot X(z, \bar{z})} e^{ik_2 \cdot X(0,0)}: \\ &= |z|^{2\alpha' k_1 \cdot k_2} :e^{ik_1 \cdot X(z, \bar{z})} e^{ik_2 \cdot X(0,0)}: \end{aligned}$$

To derive the OPE, we Taylor expand inside the normal ordering

$$:e^{ik_1 \cdot X(z, \bar{z})}::e^{ik_2 \cdot X(0,0)}: = |z|^{2\alpha' k_1 \cdot k_2} :e^{i(k_1+k_2) \cdot X(0,0)} [1 + O(z, \bar{z})]:$$

Note that OPEs (2.2.2) & (2.2.4) and so have been written asymmetrically in σ_1 & σ_2 , expanding around the latter point. They can be cast in symmetric form but the asymmetric form is ^{usually} more convenient for calculation.

2.3 Ward identities and Noether's theorem

In this section we derive general consequences of symmetry in field theory.

Consider a general field theory with action $S[\phi]$ in d spacetime dimensions, with $\phi_\alpha(\sigma)$ denoting general fields. Let there be a symmetry

$$\phi'_\alpha(\sigma) = \phi_\alpha(\sigma) + \delta\phi_\alpha(\sigma), \quad (2.3.1)$$

where $\delta\phi_\alpha(\sigma)$ is proportional to an infinitesimal parameter ϵ .

$$\Rightarrow [d\phi'] \exp(-S[\phi']) = [d\phi] \exp(-S[\phi]) \quad \Leftarrow \text{invariance}$$

A continuous symmetry in field theory implies the existence of a conserved current (Noether's theorem) & also Ward identities, which constrain the operator products of the current. To derive these, consider the change of variables,

$$\phi'_\alpha(\sigma) = \phi_\alpha(\sigma) + f(\sigma) \delta\phi_\alpha(\sigma). \quad (2.3.2) \quad (\text{Not a symmetry unless } f \text{ constant})$$

$$\Rightarrow [d\phi'] \exp[-S[\phi']] = [d\phi] \exp(-S[\phi]) \left[1 + \frac{i\epsilon}{2\pi} \int d^d\sigma g^{1/2} j^a(\sigma) \partial_a f(\sigma) + O(\epsilon^2) \right].$$

The unknown coefficient $j^a(\sigma)$ comes from the variation of the measure and the action, (both are local,) $\Rightarrow$ must be a function of the fields & their derivatives. Take the function f to be non zero only in a small region, and consider a path integral with general insertions '...' outside this region; the insertions are therefore invariant under (2.3.2).

Invariance of the path integral under change of variables gives

$$0 = \int [d\phi'] \exp(-S[\phi']) \dots - \int [d\phi] \exp(-S[\phi]) \dots$$

$$= \frac{\varepsilon}{2\pi i} \int d^d \sigma g^{1/2} f(\sigma) \langle \nabla_a j^a(\sigma) \dots \rangle \quad (2.3.5)$$

where the limited support of f has allowed us to integrate by parts.

$\Rightarrow \nabla_a j^a = 0$ as an operator equation. This is Noether's theorem.

To derive Ward identity, let $f(\sigma)$ be 1 in some region R and 0 outside R . Also, include in the path integral some general local operator $A(\sigma_0)$ at a point σ_0 inside R , and the usual arbitrary insertions '...' outside. Proceeding as above we get

$$\delta A(\sigma_0) + \frac{\varepsilon}{2\pi i} \int_R d^d \sigma g^{1/2} \nabla_a j^a(\sigma) A(\sigma_0) = 0.$$

Equivalently,

$$\nabla_a j^a A(\sigma_0) = g^{-1/2} \delta^d(\sigma - \sigma_0) \frac{2\pi}{i\varepsilon} \delta A(\sigma_0) + \text{total } \sigma\text{-derivative}$$

The divergence theorem gives $\int_{\partial R} dA n_a j^a A(\sigma_0) = \frac{2\pi}{i\varepsilon} \delta A(\sigma_0)$, with dA area element and n_a the outward normal.

Going to two flat dimensions this becomes:

$$\oint_{\partial R} (j dz - \tilde{j} d\bar{z}) A(z_0, \bar{z}_0) = \frac{2\pi}{\varepsilon} \delta A(z_0, \bar{z}_0) \quad (2.3.10)$$

$$j \equiv j_z, \quad \tilde{j} \equiv j_{\bar{z}}.$$

Important: Noether's theorem & Ward identity are local properties, so don't depend on boundary conditions & invariance under symmetry

Q: Why the change of variables (2.3.3)? Why can we just set ^{/take} f to be 0 or $\neq$ non-zero in a small region.

In conformally invariant theories ~~is~~ usually j_z is holomorphic and $j_{\bar{z}}$ anti-holomorphic (except for singularities at the other fields). In this case $(j_z, 0)$ and $(0, j_{\bar{z}})$ are separately conserved. The integral (2.3.10) then picks out the residues in OPE,

$$\text{Res}_{z \rightarrow z_0} j(z) A(z_0, \bar{z}_0) + \overline{\text{Res}}_{\bar{z} \rightarrow \bar{z}_0} \tilde{j}(\bar{z}) A(z_0, \bar{z}_0) = \frac{1}{i\varepsilon} \delta A(z, \bar{z}_0). \quad (2.3.11)$$

Here ' Res ' and ' $\overline{\text{Res}}$ ' are the coefficients of $(z-z_0)^{-1}$ and $(\bar{z}-\bar{z}_0)^{-1}$ respectively. This form of Ward identity is particularly convenient

Example; let's return to free massless scalar and consider the spacetime translation $\delta X^\mu = \epsilon a^\mu$. Under $\delta X^\mu(\sigma) = \epsilon f(\sigma) a^\mu$,

$$\delta S = \frac{\epsilon a_\mu}{2\pi\alpha'} \int d\sigma \partial^\mu X^\mu \partial_\mu \rho$$

This is of the claimed form (2.3.5) with Noether current $a_\mu j_a^\mu$, where

$$j_a^\mu = \frac{i}{\alpha'} \partial_a X^\mu$$

The components are holomorphic and antiholomorphic as expected. For the OPE of this current with the exponential operator one finds

$$j^\mu(z) : e^{ik \cdot X(0,0)} : \sim \frac{k^\mu}{2z} : e^{ik \cdot X(0,0)} : \\ \tilde{j}^\mu(\bar{z}) : e^{ik \cdot X(0,0)} : \sim \frac{k^\mu}{2\bar{z}} : e^{ik \cdot X(0,0)} :$$

from terms with single contraction. This OPE is in agreement with the general identity (2.3.11),

Another example is the world-sheet translation $\delta \sigma^a = \epsilon v^a$, under which $\delta X^\mu = -\epsilon v_a \partial^a X^\mu$. The Noether current is

$$j_a = i v^b T_{ab}, \quad (2.3.15a)$$

$$T_{ab} = -\frac{1}{\alpha'} : \left(\partial_a X^\mu \partial_b X_\mu - \frac{1}{2} \delta_{ab} \partial_c X^\mu \partial^c X_\mu \right) : \quad (2.3.15b)$$

Here T_{ab} is the world-sheet energy-momentum tensor.

2.4 Conformal invariance

The EM-tensor (2.3.15b) is traceless, $T_a^a = 0 \Leftrightarrow T_{z\bar{z}} = 0$

Conservation $\partial^a T_{ab} = 0 \Rightarrow$ in any theory with $T_a^a = 0$,

$$\bar{\partial} T_{z\bar{z}} = \partial T_{\bar{z}\bar{z}} = 0 \Rightarrow T(z) = T_{zz}(z), \quad \tilde{T}(\bar{z}) = T_{\bar{z}\bar{z}}(\bar{z}) \quad \text{are} \\ \text{respectively holomorphic and antiholomorphic.}$$

For free massless scalar,

$$T(z) = -\frac{1}{\alpha'} : \partial X^\mu \partial X_\mu : , \quad \tilde{T}(\bar{z}) = -\frac{1}{\alpha'} : \bar{\partial} X^\mu \bar{\partial} X_\mu :$$

Tracelessness of T_{ab} implies a much larger symmetry. The currents

$$j(z) = i v(z) T(z) , \quad \tilde{j}(\bar{z}) = i v(z)^* \tilde{T}(\bar{z})$$

are conserved for any holomorphic $v(z)$. For free scalar theory, OPE is

$$T(z) X^\mu(0) \sim \frac{1}{z} \partial X^\mu(0) , \quad \tilde{T}(\bar{z}) X^\mu(0) \sim \frac{1}{\bar{z}} \bar{\partial} X^\mu(0)$$

Ward identity then gives the transformation

$$\delta X^\mu = -\varepsilon v(z) \partial X^\mu - \varepsilon v(z)^* \bar{\partial} X^\mu .$$

This is an infinitesimal coordinate transformation $z' = z + \varepsilon v(z)$. The finite transformation is $\mathcal{R}$

$$X'^\mu(z', \bar{z}') = X^\mu(z, \bar{z}) , \quad z' = f(z)$$

for any holomorphic $f(z)$. This is known as conformal transformation.

Don't confuse this with diff invariance of GR. We are in flat space theory with no independent metric to vary, so transformation $z \rightarrow z'$ actually changes the distance between the points.

Conformal invariance and the OPE

Conformal invariance puts strong constraints on the form of the OPE and in particular on the OPEs of the energy-momentum tensor. Consider the OPE of T with the general operator A . Because $T(z)$ and $\tilde{T}(\bar{z})$ are (anti)holomorphic except at insertions, the corresponding coefficient functions must also have this property. $\Rightarrow$ OPE of T with general A is therefore a Laurent expansion, in integer but possibly negative powers of z . Furthermore, all singular terms are determined by the conformal transformation of A .

To see this, let us write a general expansion of the singular terms,

$$T(z) A(0,0) \sim \sum_{n=0}^{\infty} \frac{1}{z^{n+1}} A^{(n)}(0,0)$$

and similarly for $\tilde{T}$; the operator coefficients $A^{(n)}$ remain to be determined.

Under infinitesimal conformal transformation $z' = z + \epsilon V(z)$, a single pole in $V(z) T(z) A(0,0)$ arises when z^{-n-1} term of TA OPE multiplies the term of order z^n in $V(z)$. Thus the Ward identity in the form (2.3.11) implies that

$$\delta A(z, \bar{z}) = -\epsilon \sum_{n=0}^{\infty} \frac{1}{n!} \left[\partial^n V(z) A^{(n)}(z, \bar{z}) + \bar{\partial}^n V(z) \bar{A}^{(n)}(z, \bar{z}) \right].$$

Thus the operators $A^{(n)}$ are determined by the conformal transformation of A

It is convenient to take a basis of local operators that are eigenstates under

$$z' = \zeta z \quad (2.4.9), \quad (\text{a special case of conformal transformation})$$

$$\Rightarrow A'(z', \bar{z}') = \zeta^{-h} \bar{\zeta}^{-\tilde{h}} A(z, \bar{z}) \quad (2.4.13), \quad (h, \tilde{h}) \text{ are weights of } A.$$

$h + \tilde{h}$ is dimension of A (determines behaviour under scaling), $h - \tilde{h}$ is spin.

∂_z increases h by 1, $\partial_{\bar{z}}$ increases $\tilde{h}$ by one. The Ward identity & the translation $\delta A = -\epsilon V^a \partial_a A$ determine part of the OPE

$$T(z) A(0,0) = \dots + \frac{h}{z^2} A(0,0) + \frac{1}{z} \partial A(0,0) + \dots \quad (2.4.14)$$

and similarly for $\bar{T}$.

Important special case: tensor operator or primary field $\mathcal{O}$, on which general conformal transformation acts as

$$\mathcal{O}'(z, \bar{z}) = (\partial_z z')^{-h} (\partial_{\bar{z}} \bar{z}')^{-\tilde{h}} \mathcal{O}(z, \bar{z})$$

The OPE (2.4.11) reduces to

$$T(z) \mathcal{O}(0,0) = \frac{h}{z^2} \mathcal{O}(0,0) + \frac{1}{z} \partial \mathcal{O}(0,0) + \dots$$

the more singular terms in the general OPE (2.4.14) being absent.

Example: The weights of some typical operators in the free X^μ CFT:

$$\left. \begin{aligned} &X^\mu(0,0), \quad \partial X^\mu(1,0), \quad \bar{\partial} X^\mu(0,1), \quad \partial^2 X^\mu(2,0) \\ &:e^{ik \cdot X}: \quad \left(\frac{\alpha' k^2}{4}, \frac{\alpha' k^2}{4} \right) \end{aligned} \right\} \quad (2.4.17)$$

∇ All transform as tensor except $\partial^2 X^\mu$. More generally, an exponential times a general product of derivatives

$$: (\pi_i \partial^{m_i} X^{\mu_i}) (\pi_j \bar{\partial}^{n_j} X^{\nu_j}) e^{ik \cdot X} :$$

has weight

$$\left(\frac{\alpha' k^2}{4} + \sum_i m_i, \frac{\alpha' k^2}{4} + \sum_j n_j \right)$$

For any pair of operators, applying rigid translations, scale transformations and rotation to both sides of the OPE determines the z -dependence of the coefficient functions completely,

$$A_i(z_1, \bar{z}_1) A_j(z_2, \bar{z}_2) = \sum_k z_{12}^{h_k - h_i - h_j} \bar{z}_{12}^{\tilde{h}_k - \tilde{h}_i - \tilde{h}_j} c_{ij}^k A_k(z_2, \bar{z}_2),$$

where c_{ij}^k are constants. More general conformal transformations put further restrictions on the OPE: they determine the OPEs of all fields in terms of those of the Primary fields.

Conformal properties of the energy-momentum tensor

The OPE of the energy momentum tensor with itself is (using results of (2.2.11))

$$T(z) T(0) = \frac{\eta_{\mu\nu}^2}{2z^4} - \frac{2}{z^2} : \partial X^\mu(z) \partial X_\mu(0) : + : T(z) T(0) :$$

$$\sim \frac{D}{2z^2} + \frac{2}{z^2} T(0) + \frac{1}{z} \partial T(0) \quad (\text{Similar result for } \tilde{T} \text{ too}) \quad (2.4.22)$$

The OPE $T(z) \tilde{T}(\bar{z})$ must be non singular, because it can't have poles in $(z-z')$ is antiholomorphic ^{in z'} & can't have poles in $(\bar{z}-\bar{z}')$ because it is holomorphic in $\bar{z}$ (both case at non-zero separation). Same holds for any OPE between a holomorphic and an antiholomorphic operator.

$\Rightarrow T$ is not a tensor and (2.4.22) implies the transformation law

$$\mathcal{E}^{-1} \delta T(z) = -\frac{D}{12} \partial_z^3 V(z) - 2 \partial_z V(z) T(z) - V(z) \partial_z T(z),$$

In a general CFT, $T(z)$ transforms as

$$\mathcal{E}^{-1} \delta T(z) = -\frac{C}{12} \partial_z^3 V(z) - 2 \partial_z V(z) T(z) - V(z) \partial_z T(z), \quad (2.4.24)$$

Where C is a constant known as the central charge. The central charge of a free scalar is 1; for D free scalars it is D .

Note: "nontensor" refers to conformal transformations. The energy-momentum tensor will have its usual tensor property under coordinate transformations.

2.5 Free CFT

This section: discussion of three families of free-field CFTs — the linear dilation, bc, and $\beta\gamma$ theories.

Linear dilation CFT

- Family of CFTs based on the same action (2.1.10) : $S = \frac{1}{2\pi\alpha'} \int d^2z \partial X^\mu \bar{\partial} X_\mu$ but with different Energy-Momentum tensor:

$$T(z) = -\frac{1}{\alpha'} : \partial X^\mu \partial X_\mu : + V_\mu \partial^2 X^\mu$$

$$\bar{T}(z) = -\frac{1}{\alpha'} : \bar{\partial} X^\mu \bar{\partial} X_\mu : + V_\mu \bar{\partial}^2 X^\mu, \quad \text{where } V_\mu \text{ is some fixed D-vector}$$

- TT OPE is of form (2.4.25) : $T(z)T(0) \sim \frac{c}{2z^4} + \frac{2}{z^2} T(0) + \frac{1}{z} \partial T(0)$ but with central charge $c = \tilde{c} = D + 6\alpha' V_\mu V^\mu$

- TX^μ OPE & Ward identity imply the conformal transformation.

$$\delta X^\mu = -\varepsilon v \partial X^\mu - \varepsilon v^\mu \bar{\partial} X^\mu - \frac{\varepsilon}{2} \alpha' V^\mu [\partial v + (\partial v)^\dagger]$$

$\Rightarrow$ Different conformal symmetry of same action.

$\Rightarrow X^\mu$ no longer transforms as a tensor.

- Different values of V_μ are to be regarded as different CFTs.

- Is not Lorentz invariant.

bc CFT

- Has anticommuting fields b and c with action $S = \frac{1}{2\pi} \int d^2z b \bar{\partial} c$
- conformally invariant for b & c transforming as tensors of weights $(h_b, 0)$, $(h_c, 0)$:
 $h_b = \lambda$, $h_c = 1 - \lambda$, for any given constant λ .
- Family of CFT (secretely same as the linear dilation family).
- Operator eq. of motion (obtained by same method (2.1.15), (2.1.18) as before) are:
$$\begin{cases} \bar{\partial} c(z) = \bar{\partial} b(z) = 0 \\ \bar{\partial} b(z) c(0) = 2\pi \delta^2(z, \bar{z}) \end{cases}$$

- The bb & cc OPEs satisfy the EOM without source.

$$:b(z_1)c(z_2): = b(z_1)c(z_2) - \frac{1}{z_{12}}$$

- Operator products are: $b(z_1)c(z_2) \sim \frac{1}{z_{12}}$, $c(z_1)b(z_2) \sim \frac{1}{z_{12}}$

↳ double sign flip

- other operator products are non singular: $b(z_1)b(z_2) = O(z_{12})$, $c(z_1)c(z_2) = O(z_{12})$

⇒ There are not only holomorphic but have zero due to antisymmetry.

- Noether's theorem gives: $T(z) = :(\partial b)c: - \lambda \partial (:bc:)$, $\tilde{T}(\bar{z}) = 0$

- The TT OPE is of standard form (2.4.16) with $c = -3(2\lambda-1)^2 + 1$, $\tilde{c} = 0$
⇒ purely holomorphic CFT.

- Corresponding antiholomorphic theory exists: $S = \frac{1}{2\pi} \int d^2z \tilde{b} \partial \tilde{c}$
⇒ Same as the above, but with $z \leftrightarrow \bar{z}$

The bc theory has a ghost number symmetry $\delta b = -i\epsilon b$, $\delta c = i\epsilon c$. The corresponding Noether current is $j = -:bc:$. (Components are separately holo/antiholomorphic bc fields, ghost numbers are separately conserved)

This current is not a tensor:

$$T(z)j(0) \sim \frac{1-2\lambda}{z^3} + \frac{1}{z^2} j(0) + \frac{1}{z} \partial j(0)$$

⇒ Implies:

$$\epsilon^{-1} \delta j = -v \partial j - j \partial v + \frac{2\lambda-1}{2} \partial^2 v,$$

⇒ finite form:

$$(\partial_z z') j_{z'}(z') = j_z(z) + \frac{2\lambda-1}{2} \frac{\partial_z^2 z'}{\partial_z z'}$$

The one case where $h_b = h_c = \frac{1}{2} \Rightarrow c=1$, we use notation $b \rightarrow \psi$, $c \rightarrow \bar{\psi}$.

⇒ We can split bc CFT in two in a conformally invariant way

$$\psi = 2^{-1/2} (\psi_1 + i\psi_2), \quad \bar{\psi} = 2^{-1/2} (\psi_1 - i\psi_2),$$

$$S = \frac{1}{4\pi} \int d^2z (\psi_1 \bar{\partial} \psi_1 + \psi_2 \bar{\partial} \psi_2), \quad T = -\frac{1}{2} \psi_1 \partial \psi_1 - \frac{1}{2} \psi_2 \partial \psi_2$$

⇒ Each ψ theory has central charge $\frac{1}{2}$.

The bc theory for $\lambda=2$, weights $(h_b, h_c) = (2, -1)$, will arise in next chapter as the Faddeev-Popov ghosts from gauge fixing the Polyakov string.

The ψ theory will appear extensively in the more general string theories of volume two.

BR CFT

This is much like bc theory, but with commuting fields: β is an $(h_\beta, 0)$ tensor and γ an $(h_\gamma, 0)$ tensor, where $h_\beta = \lambda$, $h_\gamma = 1 - \lambda$.

Action is:

$$\Rightarrow S = \frac{1}{2\pi} \int d^2z \beta \bar{\partial} \gamma$$

Holomorphic fields: $\bar{\partial} \gamma(z) = \bar{\partial} \beta(z) = 0$ (EOM)

The EOM & operator products are derived in the standard way. Statistics are changed $\Rightarrow$ some signs in operator products are different,

$$\beta(z_1) \gamma(z_2) \sim -\frac{1}{z_{12}}, \quad \gamma(z_1) \beta(z_2) \sim \frac{1}{z_{12}}$$

EM-tensor: $T = :(\partial\beta)\gamma: - \lambda \partial(:\beta\gamma:), \quad \tilde{T} = 0$

$$\Rightarrow C = 3(2\lambda - 1)^2 - 1, \quad \tilde{C} = 0$$

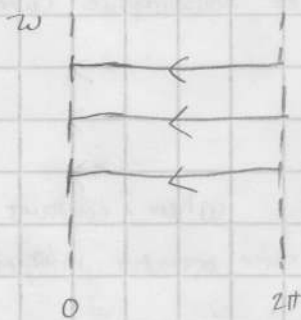
The $\beta\gamma$ -theory for $\lambda = \frac{3}{2}$, weights $(h_\beta, h_\gamma) = (\frac{3}{2}, -\frac{1}{2})$, will arise in chapter 10 as Faddeev-Popov ghosts from gauge fixing the superstring.

2.6 The Virasoro algebra

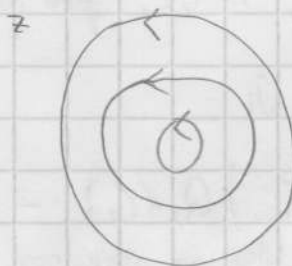
Motivation: interest in spectrum of theory

Let's first look at the closed string: $\sigma^1 \sim \sigma^1 + 2\pi$ (periodic spatial)
Euclidian time: $-\infty < \sigma^2 < \infty$, so that two dim. form infinite cylinder. ^(world)

Complex coord useful, and there are two natural choices: $w = \sigma^1 + i\sigma^2$
and $z = \exp(-iw) = \exp(-i\sigma^1 + \sigma^2)$. $\Rightarrow w \sim w + 2\pi$



a)



b)

Fig 2.3. Closed string coordinates. (a) Equal time contours in the w -plane. The dashed lines are identified. (b) Same contours in the z -plane.

The coordinates w & z are related by a conformal transformation.

For holomorphic or antiholomorphic operator we can make a Laurent expansion,

$$T_{zz}(z) = \sum_{m=-\infty}^{\infty} \frac{L_m}{z^{m+2}}, \quad \tilde{T}_{\bar{z}\bar{z}}(\bar{z}) = \sum_{m=-\infty}^{\infty} \frac{\tilde{L}_m}{\bar{z}^{m+2}}$$

The Laurent coefficients, known as Virasoro generators, are given by the contour integrals

$$L_m = \oint_C \frac{dz}{2\pi i} z^{m+2} T_{zz}(z), \quad \text{where } C \text{ is any contour encircling the origin counterclockwise.}$$

Laurent expansion = same as an ordinary Fourier transformation in the w frame at time $\sigma^2 = 0$.

$$T_{ww}(w) = - \sum_{m=-\infty}^{\infty} \exp(im\sigma^1 - m\sigma^2) T_m, \quad \tilde{T}_{\bar{w}\bar{w}}(\bar{w}) = - \sum_{m=-\infty}^{\infty} \exp(-im\sigma^1 - m\sigma^2) \tilde{T}_m,$$

where $T_m = L_m - \delta_{m,0} \frac{c}{24}$, $\tilde{T}_m = \tilde{L}_m - \delta_{m,0} \frac{\tilde{c}}{24}$. The additive shift of L_0 is from nontensor transformation (2.4.26),

$$T_{ww} = (\partial w)^2 T_{zz} + \frac{c}{24}.$$

The Hamiltonian of time translation in the $w = \sigma^1 + i\sigma^2$ frame is

$$H = \int_0^{2\pi} \frac{d\sigma^1}{2\pi} T_{22} = L_0 + \tilde{L}_0 - \frac{c + \tilde{c}}{24}.$$

Cutting open the path integral (Appendix for more info) on circles of constant time $\text{Im } w = \ln|z|$, the Virasoro generators become operators in the ordinary sense.

Holomorphy $\Rightarrow$ Integrals (2.6.6) independent of $C \Rightarrow$ invariant under time translation
 $\rightarrow$ They are conserved charges associated with the conformal transformations.

Important fact: OPE of currents determine the algebra of the corresponding charges.

Consider general charges Q_i ; $i=1,2$, given as contour integrals of holomorphic currents,

$$Q_i \{C\} = \oint_C \frac{dz}{2\pi i} j_i.$$

Consider combination $Q_1 \{C_1\} Q_2 \{C_2\} - Q_1 \{C_3\} Q_2 \{C_2\}$, where C_1, C_2, C_3 are shown in figure 2.4(a). When we slice open the path integral to make operator interpretation (operator ordering determined by time ordering, here $t_1 > t_2 > t_3$) we get

$$\hat{Q}_1 \hat{Q}_2 - \hat{Q}_2 \hat{Q}_1 = [\hat{Q}_1, \hat{Q}_2].$$

Now, for a given point z_2 on the contour C_2 , we can deform the difference of C_1 & C_3 contours as shown in figure 2.4(b), so the commutator is given by the residue of

the OPE,

$$[Q_1, Q_2] \{C_2\} = \oint_C \frac{dz_2}{2\pi i} \text{Res}_{z_1 \rightarrow z_2} j_1(z_1) j(z_2)$$

⇒ Emphasis: For conserved currents, knowing the singular terms in OPE is equivalent to knowing the commutator algebra of the corresponding charges.

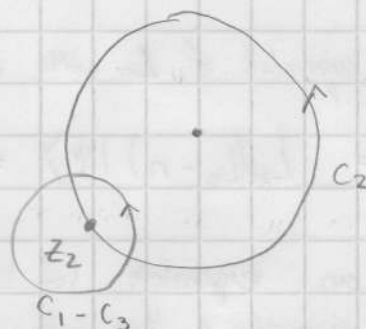
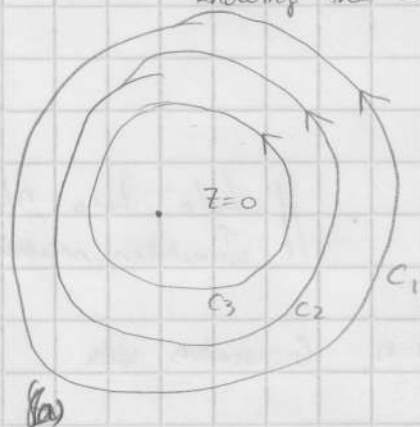


Fig. 2.4. (a) Contours centered on $z=0$. (b) For given z_2 on contour C_2 , contour $C_1 - C_3$ is contracted.

The calculation of figure 2.4 also applies with the conserved charge $Q_2 \{C_2\}$ replaced by any operator,

$$[Q, A(z_1, \bar{z}_1)] = \text{Res}_{z_1 \rightarrow z_2} j(z_1) A(z_2, \bar{z}_2) = \frac{1}{i\epsilon} \delta A(z_2, \bar{z}_2)$$

⇒ Familiar sentiment: Q generates the corresponding transformation δ .

For the contour integral of an antiholomorphic current the Ward identity & contour argument imply

$$\tilde{Q} \{C\} = - \oint_C \frac{d\bar{z}}{2\pi i} \tilde{j},$$

$$[\tilde{Q}, A(z_1, \bar{z}_1)] = \overline{\text{Res}_{\bar{z}_1 \rightarrow \bar{z}_2}} \tilde{j}(\bar{z}_1) A(z_2, \bar{z}_2) = \frac{1}{i\epsilon} \delta A(z_2, \bar{z}_2)$$

Applying this to Virasoro generators (2.6.6), where $j_m(z) = z^{m+1} T(z)$:

$$\begin{aligned} \text{Res}_{z_1 \rightarrow z_2} z_1^{m+1} T(z_1) z_2^{n+1} T(z_2) &= \text{Res}_{z_1 \rightarrow z_2} z_1^{m+1} z_2^{n+1} \left(\frac{c}{2z_{12}} + \frac{z_1}{z_{12}^2} T(z_2) + \frac{1}{z_{12}} \partial T(z_2) \right) \\ &= \frac{c}{12} (\partial^3 z_2^{m+1}) z_2^{n+1} + 2 (\partial z_2^{m+1}) z_2^{n+1} T(z_2) + z_2^{m+n+2} \partial T(z_2) \\ &= \frac{c}{12} (m^3 - m) z_2^{m+n-1} + (m-n) z_2^{m+n+1} T(z_2) + \text{total derivative} \end{aligned}$$

The contour integral of the right-hand side then gives the Virasoro algebra:

$$[L_m, L_n] = (m-n) L_{m+n} + \frac{c}{12} (m^3 - m) \delta_{m,-n}$$

The $\tilde{L}_m$ satisfy the same algebra with central charge $\tilde{c}$.

⇒ Any CFT has an infinite set of conserved charges, the Virasoro generators, which act in the Hilbert space and which satisfy the Virasoro algebra.

Let us look at some simple properties. Generally we work with eigenstates of L_0 and $\tilde{L}_0$. The generator L_0 satisfies

$$[L_0, L_n] = -n L_n.$$

If $|\psi\rangle$ is an eigenstate of L_0 with eigenvalue h , then

$$L_0 L_n |\psi\rangle = L_n (L_0 - n) |\psi\rangle = (h - n) L_n |\psi\rangle$$

$$\begin{aligned} // & L_0 L_n = L_n L_0 - n L_n \\ & \uparrow \\ & \text{commutation relation} \end{aligned}$$

⇒ $L_n |\psi\rangle$ is an eigenstate (of L_0) with eigenvalue $h - n$. Generators with $n < 0$ raise the L_0 eigenvalue & those with $n > 0$ lower it.

The three generators L_0 and $L_{\pm 1}$ form a closed algebra without central charge,

$$[L_0, L_1] = -L_1, \quad [L_0, L_{-1}] = L_{-1}, \quad [L_1, L_{-1}] = 2L_0$$

⇒ This is $SL(2, \mathbb{R})$, which differs from $SU(2)$ by signs.

For the Laurent coefficients of a holomorphic tensor field $\mathcal{O}$ of weight $(h, 0)$,

$$\mathcal{O}(z) = \sum_{m=-\infty}^{\infty} \frac{\mathcal{O}_m}{z^{m+h}}, \quad \text{one finds from OPE (2.4.16)}$$

$$\text{The commutator} \quad [L_m, \mathcal{O}_n] = [(h-1)m - n] \mathcal{O}_{m+n} \quad \begin{aligned} // & \text{Again:} \\ & n > 0 \Rightarrow \text{reduce } L_0 \\ & n < 0 \Rightarrow \text{increase } L_0 \end{aligned}$$

OPEN STRING: let

$$0 \leq \text{Re } w \leq \pi \quad \Leftrightarrow \quad \text{Im } z \geq 0,$$

where $z = -\exp(-iw)$. These coordinate regions shown in figure 2.5. At boundary:

$$T_{ab} n^a t^b = 0, \quad \text{where } n^a \text{ and } t^b \text{ are again normal \& tangent vectors.}$$

Also $T_{ab} t^a$ is conserved current so boundary condition means this current cannot move through boundary.

$$\Rightarrow T_{ww} = T_{\bar{w}\bar{w}}, \quad \text{Re } w = 0, \pi \quad \Leftrightarrow \quad T_{zz} = T_{\bar{z}\bar{z}}, \quad \text{Im } z = 0$$

it is convenient to use doubling trick. Define $T_{\bar{z}\bar{z}}$ in the lower half- z -plane as the value of $T_{\bar{z}\bar{z}}$ at its image in the upper half- z -plane, $z' = \bar{z}$:

$$T_{\bar{z}\bar{z}}(z) \equiv T_{\bar{z}\bar{z}}(\bar{z}'), \quad \text{Im } z < 0$$

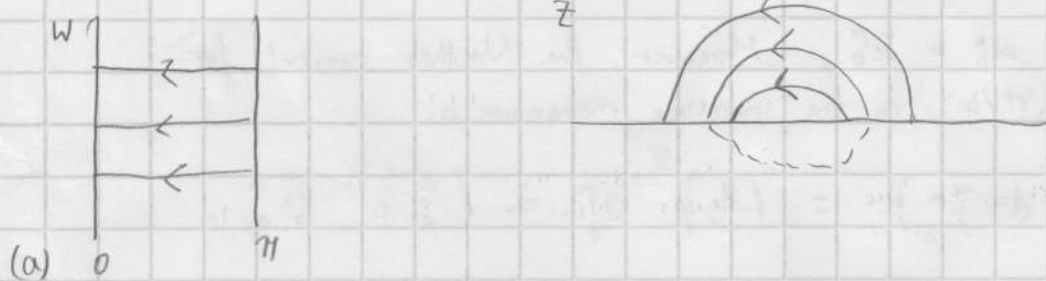


Fig 2.5. Open string coordinates. (a) Equal time contours in the w -plane. (b) The same contour in the z -plane. The dashed line shows the extension of one contour, as used in the doubling trick.

The EOM and boundary condition are then summarized by the statement that T_{zz} is holomorphic in the WHOLE complex plane. There is only one set of Virasoro generators, because the boundary condition couples T and $\bar{T}$,

$$L_m = \frac{1}{2\pi i} \int_C (dz z^{m+1} T_{zz} - d\bar{z} \bar{z}^{m+1} T_{\bar{z}\bar{z}})$$

$$= \frac{1}{2\pi i} \oint dz z^{m+1} T_{zz}(z)$$

In the first line, the contour C is a semi-circle centered on the origin; in the second line we used the doubling trick to write L_m in terms of a closed contour. Again, these satisfy the Virasoro algebra

$$[L_m, L_n] = (m-n) L_{m+n} + \frac{c}{12} (m^3 - m) \delta_{m,-n}.$$

2.7 Mode expansions

Free scalars

In free field theory, the fields break up into harmonic oscillators, and the spectrum and energy-momentum (EM) tensor can be given in terms of the modes. We start with the closed strings. In X^μ theory, ∂X and $\bar{\partial} X$ are (anti)holomorphic and so have Laurent expansions like that for T ,

$$\partial X^\mu(z) = -i \left(\frac{\alpha'}{2} \right)^{1/2} \sum_{m=-\infty}^{\infty} \frac{\alpha_m^\mu}{z^{m+1}}, \quad \bar{\partial} X^\mu(\bar{z}) = -i \left(\frac{\alpha'}{2} \right)^{1/2} \sum_{m=-\infty}^{\infty} \frac{\bar{\alpha}_m^\mu}{\bar{z}^{m+1}} \quad (2.7.1)$$

Equivalently

$$\alpha_m^\mu = \left(\frac{2}{\alpha'} \right)^{1/2} \oint \frac{dz}{2\pi} z^m \partial X^\mu(z),$$

$$\bar{\alpha}_m^\mu = - \left(\frac{2}{\alpha'} \right)^{1/2} \oint \frac{d\bar{z}}{2\pi} \bar{z}^m \bar{\partial} X^\mu(\bar{z}).$$

Single valuedness of $X^\mu \Rightarrow \alpha_0^\mu = \tilde{\alpha}_0^\mu$. Moreover, the Noether current for spacetime translation is $i\partial_\alpha X^\mu / \alpha'$, so the spacetime momentum is

$$p^\mu = \frac{1}{2\pi\alpha'} \oint_C (dz j^\mu - d\bar{z} \tilde{j}^\mu) = \left(\frac{z}{\alpha'}\right)^{1/2} \alpha_0^\mu = \left(\frac{z}{\alpha'}\right)^{1/2} \tilde{\alpha}_0^\mu$$

integrating (2.7.1) gives

$$X^\mu(z, \bar{z}) = x^\mu - i \frac{\alpha'}{2} p^\mu \ln|z|^2 + i \left(\frac{\alpha'}{2}\right)^{1/2} \sum_{\substack{m=-\infty \\ m \neq 0}}^{\infty} \frac{1}{m} \left(\frac{a_m^\mu}{z^m} + \frac{\tilde{a}_m^\mu}{\bar{z}^m} \right)$$

Either from canonical commutation, or from the contour argument & the XX OPE, we get

$$[a_m^\mu, a_n^\nu] = [\tilde{a}_m^\mu, \tilde{a}_n^\nu] = m \delta_{m,-n} \eta^{\mu\nu}$$

$$[X^\mu, p^\nu] = i\eta^{\mu\nu},$$

with other commutators vanishing.

The spectrum is given by starting with a state $|0; k\rangle$ that has momentum k^μ and is annihilated by all of the lowering modes, a_n^μ for $n > 0$, and acting in all possible ways with the raising ($n < 0$) modes.

We want to expand Virasoro generators in terms of the mode operators. Insert the Laurent expansion for X^μ into the energy-momentum tensor (2.4.4) and collect terms with a given power of z , giving

$$L_m \sim \frac{1}{2} \sum_{n=-\infty}^{\infty} \alpha_{m-n}^\mu \alpha_{\mu n} \quad (2.7.6)$$

The $\sim$ indicates that we have ignored ordering of operators. This only matters for $m=0$ (because mode operators in each term commute). For $m=0$ we put lowering operator on the right and introduce normal ordering constant,

$$L_0 = \frac{\alpha' p^2}{4} + \sum_{n=1}^{\infty} (\alpha_{-n}^\mu \alpha_{\mu n}) + A^X.$$

$\Rightarrow L_0$ is finite and unambiguous in terms of Laurent coefficients of the $::$ -ordered EM-tensor $\Rightarrow$ normal ordering constant is finite and calculable.

Using Virasoro algebra we get

$$2L_0|0; 0\rangle = (L_1 L_{-1} - L_{-1} L_1)|0; 0\rangle = 0, \quad \text{and so } A^X = 0.$$

New notation: the symbol $:\!:\!:$ will denote creation-annihilation normal ordering, placing lowering operators to the right of all raising operator, with a minus sign whenever anticommuting operators are switched. For the purposes of this definition, we will include p^μ with the lowering operators and x^μ with the raising operators

$$\Rightarrow L_m = \frac{1}{2} \sum_{n=-\infty}^{\infty} :\!:\! \alpha_{m-n}^\mu \alpha_{\mu n} :\!:\! , \quad \text{since } A^X = 0$$

⇒ We have now two forms of normal ordering: conformal normal ordering: (this will be referred to with 'normal ordering') and creation-annihilation normal ordering $::$. Let's work out relation between them. We start by comparing time-ordered and creation-annihilation-ordered product.

For the product $X^\mu(z, \bar{z}) X^\nu(z', \bar{z}')$ with $|z| > |z'|$, insert the mode expansions and move the lowering operators in $X^\mu(z, \bar{z})$ to the right. Keeping track of the commutators gives

$$X^\mu(z, \bar{z}) X^\nu(z', \bar{z}') = :: X^\mu(z, \bar{z}) X^\nu(z', \bar{z}') :: + \frac{\alpha'}{2} \eta^{\mu\nu} \left[-\ln|z|^2 + \sum_{m=1}^{\infty} \frac{1}{m} \left(\frac{z'^m}{z^m} + \frac{\bar{z}'^m}{\bar{z}^m} \right) \right]$$

$$(2.7.11) \quad = :: X^\mu(z, \bar{z}) X^\nu(z', \bar{z}') :: - \frac{\alpha'}{2} \eta^{\mu\nu} \ln|z - z'|^2$$

Since $|z| > |z'|$, the left-hand side here is time-ordered and the sum converges. The definition (2.1.21) gave the relation between the time-ordered product and the conformal-normal-ordered product and the relation is the same as the relation (2.7.11), so that

$$:: X^\mu(z, \bar{z}) X^\nu(z', \bar{z}') :: = : X^\mu(z, \bar{z}) X^\nu(z', \bar{z}') :$$

- This doesn't hold in general in CFT. (The arbitrary grouping of p^μ with lowering operators was done to get this result.)
- Also doesn't hold for operators conformal-normal-ordered in terms of w rather than z , for example.

Creation-annihilation normal products of more than two X^μ 's have same combinatoric properties as for $::$ -ordering. They follow Wick's theorem from field theory.

To convert operators normal ordered in one form to another normal ordering, one sums over all subtraction using the difference of the two point functions. That is, if we have two kinds of ordering,

$$[X^\mu(z, \bar{z}) X^\nu(z', \bar{z}')]_1 = [X^\mu(z, \bar{z}) X^\nu(z', \bar{z}')]_2 + \eta^{\mu\nu} \Delta(z, \bar{z}, z', \bar{z}'),$$

then for a general operator F ,

$$[F]_1 = \exp \left(\frac{1}{2} \int d^2z d^2z' \Delta(z, \bar{z}, z', \bar{z}') \frac{\delta}{\delta X^\mu(z, \bar{z})} \frac{\delta}{\delta X_\mu(z', \bar{z}')} \right) [F]_2.$$

For the Linear dilation CFT, the Laurent expansion and commutators are unchanged, while the Virasoro generators contain the extra term

$$L_m = \frac{1}{2} \sum_{n=-\infty}^{\infty} \alpha_{m-n}^\mu \alpha_{\mu n} + i \left(\frac{\alpha'}{2} \right)^{1/2} (m+1) V^\mu \alpha_{\mu m}$$

bc CFT

The fields b and c have the Laurent expansions

$$b(z) = \sum_{m=-\infty}^{\infty} \frac{b_m}{z^{m+\lambda}}, \quad c(z) = \sum_{m=-\infty}^{\infty} \frac{c_m}{z^{m+1-\lambda}}.$$

(There are only Laurent expansion for $\lambda = \text{integer}$, which we will assume for now.)

The OPE gives the anticommutators

$$\{b_m, c_n\} = \delta_{m, -n}$$

Consider first the states that are annihilated by all of the $n > 0$ operators.

The b_0, c_0 oscillator algebra generates two such states $|\uparrow\rangle$ and $|\downarrow\rangle$, with

$$b_0 |\downarrow\rangle = 0, \quad b_0 |\uparrow\rangle = |\downarrow\rangle, \quad c_0 |\downarrow\rangle = |\uparrow\rangle, \quad c_0 |\uparrow\rangle = 0,$$

$$b_n |\downarrow\rangle = b_n |\uparrow\rangle = c_n |\downarrow\rangle = c_n |\uparrow\rangle = 0, \quad n > 0.$$

The general state is obtained by acting on these states with $n < 0$ modes at most once each (because they anticommute). It is conventional to group b_0 with the lowering operators and c_0 with raising operators, so we will single out $|\downarrow\rangle$ as the ghost vacuum $|0\rangle$.

The Virasoro generators are

$$L_m = \sum_{n=-\infty}^{\infty} (m\lambda - n) :b_n c_{m-n}: + \delta_{m,0} a^g.$$

The ordering constant can be determined as in eq (2.7.8), which gives

$$2L_0 |\downarrow\rangle = (L_1 L_{-1} - L_{-1} L_1) |\downarrow\rangle$$

$$= (\lambda b_0 c_1) [(1-\lambda) b_{-1} c_0] |\downarrow\rangle = \lambda(1-\lambda) |\downarrow\rangle$$

$$\Rightarrow a^g = \frac{1}{2} \lambda(1-\lambda) \text{ and}$$

$$L_m = \sum_{n=-\infty}^{\infty} (m\lambda - n) :b_n c_{m-n}: + \frac{\lambda(1-\lambda)}{2} \delta_{m,0}$$

For the ghost number current (2.5, 14), $j = - :bc:$, the charge is

$$N^g = - \frac{1}{2\pi i} \int_0^{2\pi} d\omega j\omega$$

$$= \sum_{n=1}^{\infty} (c_{-n} b_n - b_{-n} c_n) + c_0 b_0 - \frac{1}{2}$$

It satisfies $[N^g, b_m] = -b_m$, $[N^g, c_m] = c_m$,

and so count the number of c minus the number of b excitations. The ground states have ghost numbers $\pm \frac{1}{2}$:

$$N^g |\downarrow\rangle = -\frac{1}{2} |\downarrow\rangle, \quad N^g |\uparrow\rangle = \frac{1}{2} |\uparrow\rangle.$$

This depends on the value of the ordering constant, but it could be guessed on the grounds that the average ghost number of the ground states should be zero: the ghost number changes sign under $b \leftrightarrow c$.

The $\beta\gamma$ theory is similar (details in chapter 10).

Open strings

In the open string, Neumann boundary condition becomes $\partial_z X^\mu = \partial_{\bar{z}} X^\mu$ on the real axis. There is only one set of modes, the boundary condition requiring $\alpha_m^\mu = \tilde{\alpha}_m^\mu$ in the expansion (2.7.1). The spacetime momentum integral (2.7.3) runs only over a semi-circle, so the normalization is now

$$\alpha_0^\mu = (2\alpha')^{1/2} p^\mu$$

The expansion for X^μ is then

$$X^\mu(z, \bar{z}) = x^\mu - i\alpha' p^\mu \ln|z|^2 + i\left(\frac{\alpha'}{2}\right)^{1/2} \sum_{\substack{m=-\infty \\ m \neq 0}}^{\infty} \frac{\alpha_m^\mu}{m} (z^{-m} + \bar{z}^{-m})$$

Also,
$$L_0 = \alpha' p^2 + \sum_{n=1}^{\infty} \alpha_{-n}^\mu \alpha_{n\mu}$$

The commutators are as before,

$$[\alpha_m^\mu, \alpha_n^\nu] = m \delta_{m,-n} \eta^{\mu\nu}, \quad [x^\mu, p^\nu] = i\eta^{\mu\nu}$$

relevant

For bc theory, the boundary conditions are $c(z) = \tilde{c}(\bar{z})$, $b(z) = \tilde{b}(\bar{z})$, $\text{Im}(z) = 0$, written in terms of coordinates z where the boundary is the real axis. We can therefore use the doubling trick to write the holomorphic & antiholomorphic fields in the upper-half plane in terms of holomorphic fields in the whole plane, $c(z) \equiv \tilde{c}(\bar{z}')$, $b(z) \equiv \tilde{b}(\bar{z}')$, $\text{Im}(z) \leq 0$, $z' = \bar{z}$.

The open string thus has a single set of Laurent modes of each of b & c .

2.8 Vertex operators

In QFT there is space of states (of the theory) and set of local operators. In CFT there is a simple and useful isomorphism between these, with the CFT quantized on a circle.

Consider the semi-infinite cylinder in the w -coordinate,

$$0 \leq \text{Re } w \leq 2\pi, \quad w \sim w + 2\pi, \quad \text{Im } w \leq 0,$$

which maps into the unit disk in the coordinate $z = \exp(-iw)$, as shown in Fig 2.6:

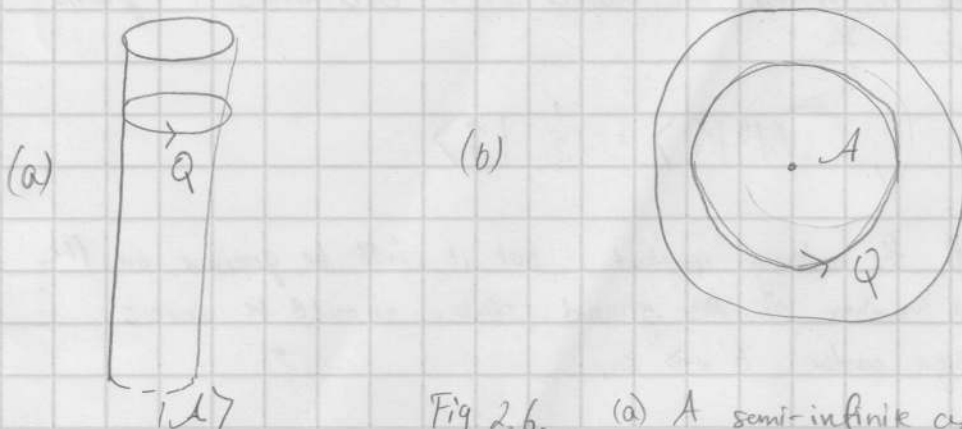


Fig 2.6. (a) A semi-infinite cylinder in w , with initial state $|A\rangle$ and charge Q . (b) The conformally equivalent unit disk with local operator A and contour integral for Q .

To define path integral in the w -coordinate one must in particular specify the boundary condition as $\text{Im } w \rightarrow -\infty$. That is one must specify the initial state. In the z -coordinate, $\text{Im } w \rightarrow -\infty$ maps to the origin, so this is equivalent to specifying the behaviour of the fields at that point. In effect, this defines a local operator at the origin, known as the vertex operator associated with the state. Going in another direction, the path integral on the disk with an operator A at the origin maps to the path integral on the cylinder with a specified initial state $|A\rangle$.

For free field theories one can easily work out the detailed form of this isomorphism. Suppose one has a conserved charge Q acting on the state $|A\rangle$ as in figure 2.6(a). One can find the corresponding local operator by using the OPE to evaluate the contour integral in figure 2.6(b). We can use this to identify state $|1\rangle$ corresponding to the unit operator.

With no operator at the origin, ∂X^m & $\bar{\partial} X^m$ are (anti)holomorphic inside the Q contour in figure 2.6(b). The contour integrals (2.7.2) defining α_m^m and $\tilde{\alpha}_m^m$ for $m \geq 0$ have no poles, and so vanish. Thus $|1\rangle$ is annihilated by these modes. This identifies it as the ground state $|1\rangle = |0; 0\rangle$, with the normalization chosen for convenience. Now consider, for example, the state $\alpha_m^m |1\rangle$ with

m positive. Passing to figure 2.6(b) with $Q = \alpha_{-m}^\mu$ surrounding the unit operator, the fields are holomorphic inside the contour, and so we can evaluate

$$\alpha_{-m}^\mu = \left(\frac{2}{\alpha'}\right)^{1/2} \oint \frac{dz}{2\pi} z^{-m} \partial X^\mu(z) \rightarrow \left(\frac{2}{\alpha'}\right)^{1/2} \frac{i}{(m-1)!} \partial^m X^\mu(0) \quad (2.8.3)$$

for $m \geq 1$. Thus

$$\alpha_{-m}^\mu |1\rangle \cong \left(\frac{2}{\alpha'}\right)^{1/2} \frac{i}{(m-1)!} \partial^m X^\mu(0), \quad m \geq 1$$

Similarly

$$\tilde{\alpha}_{-m}^\mu |1\rangle \cong \left(\frac{2}{\alpha'}\right)^{1/2} \frac{i}{(m-1)!} \bar{\partial}^m X^\mu(0), \quad m \geq 1$$

This correspondence also holds when α_{-m}^μ or $\tilde{\alpha}_{-m}^\mu$ acts on general state $|A\rangle$.

If $:A(0,0):$ is any normal-ordered operator, there may be singularities in the operator product of $\partial X^\mu(z)$ with $:A(0,0):$, but it is not hard to check that

$\alpha_{-m}^\mu :A(0,0): = : \alpha_{-m}^\mu A(0,0) :$ for $m > 0$ because the contour integral of the contractions will never have a single pole. One can then carry out the same manipulation (2.8.3) inside the normal ordering, and so the state with several α oscillators excited comes out as the $::$ normal-ordered product of the corresponding operators,

$$\alpha_{-m}^\mu \rightarrow i \left(\frac{2}{\alpha'}\right)^{1/2} \frac{1}{(m-1)!} \partial^m X^\mu(0), \quad m \geq 1 \quad (2.8.7a)$$

$$\tilde{\alpha}_{-m}^\mu \rightarrow -i \frac{1}{(m-1)!} \bar{\partial}^m X^\mu(0), \quad m \geq 1. \quad (2.8.7b)$$

$$\text{Similarly, } X_0^\mu \rightarrow X^\mu(0,0). \quad (2.8.8)$$

$\Rightarrow$ Any state can be obtained from $|1\rangle$ by acting with the operators on the left-hand sides of eqs. (2.8.7) and (2.8.8). The operator corresponding to this state is then given by the $::$ normal-ordered product of the corresponding local operators on the right-hand sides. For example,

$$|0; k\rangle \cong : e^{ik \cdot X(0,0)} :$$

$\Rightarrow$ Under translation $X^\mu \rightarrow X^\mu + a^\mu$, both the state and the corresponding operator have the same transformation, being multiplied by $\exp(ik \cdot a)$.

The same method applies to the bc theory. For clarity let's look at $\alpha=2$ case:
From Laurent expansions (2.7.16) and the contour argument it follows that

$$b_m |1\rangle = 0, \quad m \geq -1, \quad c_m |1\rangle = 0, \quad m \geq 2$$

⇒ Due to shifts in exponents of the Laurent Expansion coming from the conformal weights of b & c , $|1\rangle$ no longer maps to a ground state.

$$\Rightarrow |1\rangle = b_{-1}|\downarrow\rangle$$

The translation of raising operators is straightforward,

$$b_{-m} \rightarrow \frac{1}{(m-2)!} \partial^{m-2} b(0), \quad m \geq 2,$$

$$c_{-m} \rightarrow \frac{1}{(m+1)!} \partial^{m+1} c(0), \quad m \geq -1.$$

Notice that ghost number $-\frac{3}{2}$ state $b_{-1}|\downarrow\rangle$ maps to the ghost number 0 unit operator, and the ghost number $-\frac{1}{2}$ state $|\downarrow\rangle$ to the ghost number 1 operator c . The difference comes from the nontensor property (2.5.17) of the ghost number current

$$(\partial_z w) j_w(w) = \bar{j}_z(z) + q_0 \frac{\partial_z^2 w}{\partial_z w} = \bar{j}_z(z) - \frac{q_0}{z},$$

$$\text{where } q_0 = \lambda - \frac{1}{2} = \frac{3}{2}.$$

The ghost number of states is conventionally defined by the cylindrical frame expression N^g , eq. (2.7.22), while the contour argument of figure 2.6 relates the ghost number of the vertex operator to the radial frame charge

$$Q^g \equiv \frac{1}{2\pi i} \oint dz j_z = N^g + q_0$$

⇒ This applies to other charges as well; for a tensor current the charges of a state and corresponding operator are equal.

Most of above can be extended to $\beta\gamma$ theory (some complications arise in superstring which is why we return to this in chapter 10).

All ideas of this section also extends to open strings. The semi-infinite strip $0 \leq \text{Re } w \leq \pi$, $\text{Im } w \leq 0$, maps to a half disk, the intersection of the upper half-plane and the unit circle, under $z = -\exp(-iw)$. The initial state again maps to the origin, which is on the boundary, so there is an isomorphism

local operators on the boundary $\leftrightarrow$ states on the interval.

Details left for the reader. (Doubling trick useful once again in making contour arguments.)

Path integral derivation

Consider a unit disk in the z -plane, with local operator A at the origin and with the path integral fields ϕ fixed to some specific boundary values ϕ_0 on the unit circle. The path integral over the fields ϕ_i on the interior of the disk with ϕ_0 held fixed produces some functional $\Psi_A[\phi_0]$,

$$\Psi_A[\phi_0] = \int [d\phi_i]_{\phi_0} \exp(-S[\phi_i] A(0))$$

A functional of the fields is the Schrödinger representation of a state, so this is a mapping from operators to states.

To go the other way, start with some state $\Psi[\phi_0]$. Consider a path integral over the annular region $1 \geq |z| \geq r$, with the fields ϕ_0 on the outer circle fixed and the fields ϕ'_0 integrated over the inner circle as follows:

$$\int [d\phi'_0] [d\phi_i]_{\phi_0, \phi'_0} \exp(-S[\phi_i]) r^{-L_0 - \tilde{L}_0} \Psi[\phi'_0] \quad (2.8.18)$$

($\rightarrow$ Integral weighted by the state (functional) $r^{-L_0 - \tilde{L}_0} \Psi[\phi'_0]$.)

Now, the path integral over the annulus just corresponds to propagating from $|z|=r$ to $|z|=1$, which is equivalent to acting with the operator $r^{L_0 + \tilde{L}_0}$. This undoes the operator acting on Ψ , so the path integral (2.8.18) is again $\Psi[\phi_0]$. Now take limit $r \rightarrow 0$. The annulus becomes a disk, and the limit of the path integral over the inner circle can be thought of as defining some local operators at origin. By construction the path integral on disk with this operator reproduces $\Psi[\phi_0]$ on the boundary.

Let's work this out explicitly for a single free scalar field X . On the unit circle expand

$$X_0(\theta) = \sum_{n=-\infty}^{\infty} X_n e^{in\theta}, \quad X_n^* = X_{-n}$$

$\Rightarrow$ The boundary state $\Psi[X_0]$ can be regarded as a function of all of the X_n .

First, we identify the state corresponding to the unit operator, given by path integral without insertions

$$\Psi_1[X_0] = \int [dX_i]_{X_0} \exp\left(-\frac{1}{2\pi\alpha'} \int d^2z \partial X \bar{\partial} X\right)$$

Evaluate this by the usual Gaussian method. Separate X_i into a classical part and a fluctuation,

$$X_i = X_{cl} + X'_i$$

$$X_{ce}(z, \bar{z}) = X_0 + \sum_{n=1}^{\infty} (z^n X_n + \bar{z}^n X_{-n}).$$

With this definition, X_{ce} satisfies the equation of motion and X'_i vanishes on the boundary. The path integral then separates,

$$\Psi_1[X_b] = \exp(-S_{ce}) \int [dX'_i]_{X_b=0} \exp\left(-\frac{1}{2\pi\alpha'} \int d^2z \partial X' \bar{\partial} X'\right),$$

$$\text{with } S_{ce} = \frac{1}{2\pi\alpha'} \sum_{m,n=1}^{\infty} m n X_m X_{-n} \int_{|z|<1} d^2z z^{m-1} \bar{z}^{n-1}$$

$$= \frac{1}{\alpha'} \sum_{m=1}^{\infty} m X_m X_{-m}.$$

The X'_i integral is constant, independent of boundary condition, so

$$\Psi_1[X_b] \propto \exp\left(-\frac{1}{\alpha'} \sum_{m=1}^{\infty} m X_m X_{-m}\right). \quad (2.8.24)$$

This is Gaussian, and is in fact ground state. To see this, write the raising and lowering operators in the Schrödinger basis,

$$\alpha_n = -\frac{i n}{(2\alpha')^{1/2}} X_{-n} - i \left(\frac{\alpha'}{2}\right)^{1/2} \frac{\partial}{\partial X_n},$$

$$\tilde{\alpha}_n = -\frac{i n}{(2\alpha')^{1/2}} X_n - i \left(\frac{\alpha'}{2}\right)^{1/2} \frac{\partial}{\partial X_{-n}},$$

as follows from the Laurent expansion at $|z|=1$ and the mode algebra.

Acting on (2.8.24), we find

$$\alpha_n \Psi_1[X_b] = \tilde{\alpha}_n \Psi_1[X_b] = 0, \quad n \geq 0,$$

so this is indeed the ground state. Thus, $|1\rangle \propto |0; 0\rangle$.

We actually define $|1\rangle = |0; 0\rangle$ instead of keeping track of normalization of $\int_{\text{path}}$ integral.

Another calculation is the state corresponding to $\partial^k X$; this just adds a factor of $\partial^k X_{ce}(0) = k! X_k$ to the result, so

$$|\partial^k X_k\rangle = k! X_k \Psi_1 = -i \left(\frac{\alpha'}{2}\right)^{1/2} (k-1)! \alpha_k |0; 0\rangle,$$

and similarly for $\bar{\partial}^k X$. This extends to all products of exponentials and derivatives of X . The conformal normal ordering just cancels the effect of the X'_i path integral.

2.9 More on states and operators

The OPE

In this section we make additional applications of the state-operator correspondence.

We start with a simple and general derivation of the OPE, as shown in Figure 2.7.

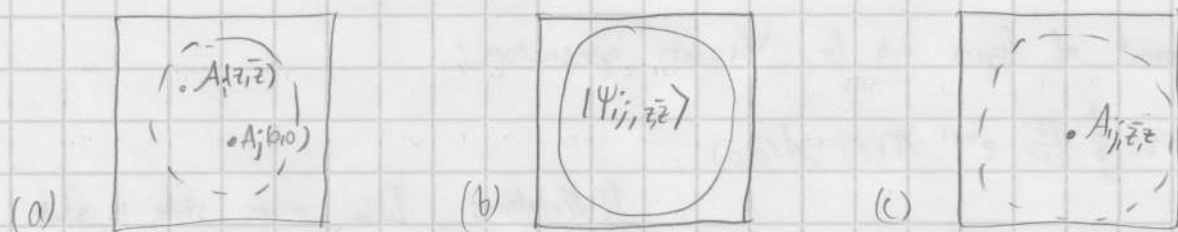


Fig 2.7 (a) World sheet with two local operators. (b) Integration over fields on the interior of the disk produces boundary state $|\Psi_{ij}, z, \bar{z}\rangle$. (c) Sewing in a disk with the corresponding local operator. Expanding in operators of definite weight gives the OPE.

Consider the product $A_i(z, \bar{z}) A_j(0,0)$, where $|z| < 1$.

We can divide the path integral into an integral over fields ϕ_i on the interior of the unit disk, an integral over fields ϕ_b on the unit circle, and an integral over fields ϕ_e on the exterior of the unit disk. (More details on cutting path integrals in Appendix of the book.)

As in the discussion at the end of the previous section, the integral over ϕ_i leaves some functional of ϕ_b ; call it $\Psi_{ij, z, \bar{z}}[\phi_b]$. By the state-operator correspondence this is equivalent to gluing back the disk with the appropriate local operator $A_{ij, z, \bar{z}}(0,0)$ and so we are done! To put this in standard form expand in a complete set of $L_0, \bar{L}_0$ eigenstates,

$$A_{ij, z, \bar{z}} = \sum_k z^{h_k - h_i - h_j} \bar{z}^{\tilde{h}_k - \tilde{h}_i - \tilde{h}_j} c_{ij}^k A_k,$$

with the z - and $\bar{z}$ -dependence determined by the conformal weights as in eq. (2.4.20). The convergence of OPE is just the usual convergence of a complete set in quantum mechanics. The construction is possible as long as there are no other operators with $|z'| \leq |z|$, so that we can cut a circle of radius $|z| + \epsilon$.

For three operators $A_i(0,0) A_j(1,1) A_k(z, \bar{z})$, the regions of convergence of the $z \rightarrow 0$ and $z \rightarrow 1$ OPEs, $|z| < 1$ and $|1-z| < 1$, overlap. The coefficient of A_m in the triple product can then be written as a sum involving $c_{ik}^l c_{lj}^m$ or as a sum involving $c_{jk}^l c_{li}^m$. (The sums are equal due to associativity.)

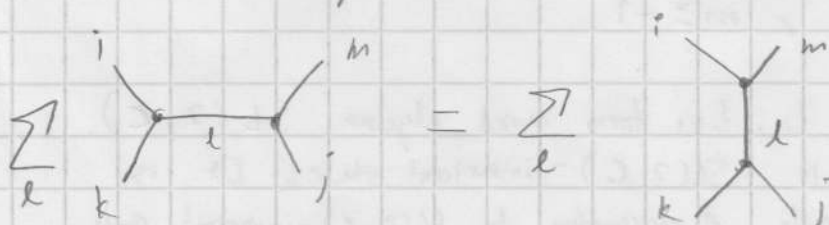


Fig 2.8 Schematic picture of OPE associativity.

The Virasoro algebra and highest weight states

Consider the argument of figure 2.6 for Virasoro generators:

$$L_m |A\rangle \equiv \oint \frac{dz}{2\pi i} z^{m+1} T(z) A(0,0)$$

$$\equiv L_m \cdot A(0,0)$$

// Notation: L_m maps state to state
 $L_m \cdot$ maps local operators to local operators

State-operator isomorphism: operator acting on Hilbert space has image acting on the space of local operators. $\Rightarrow$ Form the corresponding contour around the operator and evaluate the resulting local operator using the OPE.

In this notation TA OPE is

$$T(z) A(0,0) = \sum_{m=-\infty}^{\infty} z^{-m-2} L_m \cdot A(0,0) \quad (2.9.5)$$

This relates to earlier notation (2.4.11) via $A^{(n)} = L_{n-1} \cdot A$ for $n \geq 0$, and the conformal transformation of A is

$$\delta A(z, \bar{z}) = -\varepsilon \sum_{n=0}^{\infty} (n!)^{-1} \left[\partial^n v(z) L_{n-1} + (\partial^n \bar{v}(z))^* \tilde{L}_{n-1} \right] \cdot A(z, \bar{z}).$$

Using the general form (2.4.14) for the TA OPE, we get useful result:

$$L_{-1} \cdot A = \partial A, \quad \tilde{L}_{-1} \cdot A = \bar{\partial} A, \quad L_0 \cdot A = h A, \quad \tilde{L}_0 \cdot A = \tilde{h} A \quad (2.9.7)$$

$\Rightarrow$ The OPE (2.9.5) implies that a primary field $\mathcal{O}$ of weight $(h, \tilde{h})$ corresponds to a state satisfying

$$L_0 |\mathcal{O}\rangle = h |\mathcal{O}\rangle, \quad \tilde{L}_0 |\mathcal{O}\rangle = \tilde{h} |\mathcal{O}\rangle,$$

$$L_m |\mathcal{O}\rangle = 0 = \tilde{L}_m |\mathcal{O}\rangle, \quad m > 0$$

Such a state is known as a highest weight state. For CFTs of interest $L_0, \tilde{L}_0$ are bounded by below: by acting repeatedly with lowering operators we can reach highest weight state.

Interesting special case: unit operator, which has nonsingular operator product T 1

In any CFT $\Rightarrow L_m |1\rangle = \tilde{L}_m |1\rangle = 0, \quad m \geq -1$

As noted in section (2.6) the operators $L_0, L_{\pm 1}$ form crossed algebra $SL(2, \mathbb{C})$.

(Same for $\tilde{L}, \tilde{L}_{\pm 1}$) $\Rightarrow |1\rangle$ is $SL(2, \mathbb{C})$ -invariant state. It is only such state because (2.9.7) imply A corresponding to $SL(2, \mathbb{C})$ -invariant state is independent of position $\Rightarrow$ must be a c-number.

Unitary CFTs

Highest weight states play a special role in string theory and in the theory of representation of Virasoro algebra. Let's derive a few important general results in unitary CFT.

Unitary CFT is one that has positive inner product $\langle 1 | 1 \rangle$ such that

$$L_m^\dagger = L_{-m}, \quad \tilde{L}_m^\dagger = \tilde{L}_{-m} \quad \left| \begin{array}{l} \text{Recall the definition of adjoint} \\ \langle \alpha | A \beta \rangle = \langle A^\dagger \alpha | \beta \rangle \end{array} \right.$$

For example, the X^μ CFT is unitary for spacelike μ if we take the inner product

$$\langle 0; k | 0; k' \rangle = 2\pi \delta(k - k')$$

for the ground state and define $\alpha_m^\dagger = \alpha_{-m}$, $\tilde{\alpha}_m^\dagger = \tilde{\alpha}_{-m}$.

$\Rightarrow$ This implicitly defines inner product for all higher states. This CFT is not unitary for $\mu=0$ because of the opposite sign in the commutator there.

The first constraint for any state in a unitary CFT is that it must have $h, \tilde{h} \geq 0$. If true for highest weight state $\Rightarrow$ true for all states. Virasoro algebra for highest weight state gives

$$\begin{aligned} 2h_0 \langle 0 | 0 \rangle &= 2 \langle 0 | L_0 | 0 \rangle = \langle 0 | [L_1, L_{-1}] | 0 \rangle \\ &= \|L_{-1} | 0 \rangle\|^2 \geq 0 \end{aligned} \quad (2.9.14)$$

$\Rightarrow h_0 \geq 0$. It also follows that if $h_0 = \tilde{h}_0 = 0$ then $L_1 \cdot \mathcal{O} = \tilde{L}_1 \cdot \mathcal{O} = 0$, and so $\mathcal{O}$ is independent of position. $\Rightarrow \mathcal{O}$ is c-number. $\Rightarrow$ Unit operator is the only $(0,0)$ operator in a unitary CFT. X^μ itself is the one notable exception: the corresponding state $X^\mu |0,0\rangle$ is nonnormalizable because of the infinite range of X^μ , and eq. (2.9.14) doesn't hold. The general theorems are of most use for CFTs corresponding to compactified dimensions, where this kind of exception cannot occur.

In unitary CFT an operator is only holomorphic if and only if $\tilde{h} = 0$, and antiholomorphic if and only if $h = 0$; this is an important result

$$\bar{\partial} A = 0 \Leftrightarrow \tilde{h} = 0, \quad \partial A = 0 \Leftrightarrow h = 0$$

Finally, using the above argument with the commutator $[L_n, L_n]$, one can show that $c, \tilde{c} \geq 0$ in a unitary CFT. In fact, the only unitary CFT with $c=0$ is trivial one, $L_n = 0$.

Zero-point energies

The state-operator mapping gives a simple alternative derivation of the various normal ordering constants. In any CFT, $L_0 |1\rangle = 0$ determines the additive normalization of L_0 . In X CFT, $|1\rangle = |0; 0\rangle$, so $\alpha^\dagger$ vanishes. In bc-theory, $|1\rangle$ is the excited state $b_{-1} |0\rangle$, so the weight of $|1\rangle$ is -1 in agreement with result (2.7.21) for $\lambda=2$.

⇒ We get a physical interpretation of the central charge. In unitary CFT the ground state is $|1\rangle$ with $L_0 = \tilde{L}_0 = 0$. The conformal transformation (2.6.10) between the radial and time-translation generators then imply that

$$E = \frac{c + \tilde{c}}{24}$$

for the ground state. This is Casimir energy, coming from the finite size of the system, and it only depends on the central charge. On dimensional grounds this energy is inverse to the size l of the system, which is 2π here, so the general result would be

$$E = \frac{\pi(c + \tilde{c})}{12l}$$

We have now given three honest ways of calculating normal ordering constants:

- Virasoro algebra as in eq. (2.7.8)
- Relating the two forms of normal ordering as in (2.7.11)
- State-operator mapping above

Nevertheless the idea of adding zero-point energy is intuitive and useful mnemonic, so we give a prescription that can be checked by one of the more honest methods:

1. Add the zero-point energies, $\frac{1}{2}\omega$ for each bosonic mode and $-\frac{1}{2}\omega$ for each fermionic (anticommuting) mode.
2. One encounters divergent sums of the form $\sum_{n=1}^{\infty} (n - \theta)$, the θ arising when one considers nontrivial periodicity conditions. Define this to be
(2.9.19) $\sum_{n=1}^{\infty} (n - \theta) = \frac{1}{24} - \frac{1}{8}(2\theta-1)^2$
This is the value one obtains as in eq. (1.3.32) by regulating and discarding quadratically ^{divergent} part.
3. The above gives the normal ordering constant for the w-frame generator T_0 , eq. (2.6.8). For L_0 we must add the nontensor correction $\frac{1}{24}c$.

For free bosons, the modes are integer-valued so we get one-half of the sum (2.9.19) for $\theta=0$ after step 2, which is $-\frac{1}{24}$. This is just offset by the correction in step 3, giving $\alpha^\dagger = 0$. For the ghosts we similarly get $\frac{26}{24} - \frac{26}{24} = -1$